WHEN REUSABLE TEMPORAL SKILLS BECOME CAUSAL BOTTLENECKS IN EVOLVING TIME-SERIES AGENTS

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ABSTRACT

Agents that revisit evolving institutional data repeatedly face procedures whose inputs change while the operation itself should remain reusable. TimeSage-EV reports recurring temporal-validity and exogenous-context failures together with aggregate gains from a reusable skill library. We ask when a reusable procedure becomes an intervention-defined causal bottleneck: the targeted procedure must repair its matched failure over both a complexity-matched target-blind helper and the unassisted agent in a non-ceiling model-task cell. This definition identifies a binding repair operation; it does not claim that the procedure is the unique possible cause of failure. Our source-native study contains 35 frozen endpoints from three first-party institutional systems. The registered DeepSeek track uses five repeats on the initial BEA/NOAA replay. A provenance-clean Kimi replication retains four repeats after a controller-retry exclusion. An exploratory mini-model layer probes capacity dependence. A post-review EIA validation then adds 12 energy endpoints while reusing the cutoff skill byte-for-byte. Documentary grounding gives the cleanest primary repair: on five acquisition endpoints DeepSeek reaches 60% versus 20% for both controls. On the two registered same-domain held-out grounding endpoints, both DeepSeek and Kimi show a descriptive 50-point gain over both controls; cross-domain grounding shows no incremental effect. Cutoff filtering is regime-dependent. On eight held-out EIA endpoints, DeepSeek rises from 47.5% without the skill and 70% with the generic helper to 100% with targeted filtering, while the mini model is already at no-skill ceiling on the same tasks. Release alignment improves held-out Kimi performance on three endpoints by 41.7 points over no skill. Across 1,326 retained runtime-valid rows, the evidence supports a conditional causal account: reusable temporal procedures become binding repair operations in specific non-ceiling regimes, while transfer depends on the procedure and target regime.

1 INTRODUCTION

Evolving time-series analysis requires an agent to revisit the same scenario as new releases arrive. The evidence cutoff changes every period. Historical documents remain useful, while later information must stay hidden. TimeSage-EV formalizes this setting across 60 institutional scenarios and 1,485 scenario-period questions (Yao et al., 2026).

The benchmark reveals failures with a procedural character. Agents can misuse temporal cutoffs. They can align a release with the wrong reporting period. They can also miss documentary context needed to explain a numerical change. The concrete values change across periods, yet the operation needed to avoid the error is reusable.

TimeSage-1.0 already shows aggregate gains from a reusable skill library (Yao et al., 2026), while recent work provides paired/no-skill attribution, causal masking, continual skill evaluation, and direct skill compilation or optimization (Zhou et al., 2026; Dong et al., 2026; Wang et al., 2026b; Guan et al., 2026; Wei et al., 2026; Zheng et al., 2026; Xu et al., 2026). We therefore do not claim priority for causal skill evaluation or reusable-skill learning. Our narrower question is whether one frozen operation is a binding repair for a *predeclared recurring temporal failure*: it must beat both a matched generic helper and the original agent, and its transfer, ceiling, and domain boundaries must remain visible.

We therefore use a three-arm intervention. The same frozen endpoint is evaluated with a mechanism-specific skill. A complexity-matched target-blind helper supplies the generic control. A no-skill arm anchors the original agent. The generic helper is matched for callable interface and implementation complexity, not assumed to be behaviorally neutral. This distinction is deliberate: a generic helper can itself hurt performance, creating an apparent targeted-minus-generic gain without repair over the original agent. We therefore never call a targeted-minus-generic gain a repair unless the targeted arm also improves over no skill.

We use *causal bottleneck* as an intervention-defined term. A procedure is a bottleneck in a model-task regime when adding the frozen targeted procedure corrects the registered family failure over both controls on the same endpoint. The claim is about a binding repair operation under this intervention. It does not require the procedure to be the only possible intervention that could improve the agent. This distinction lets the paper make a causal repair claim without assuming causal exclusivity.

To isolate the mechanism on content-addressed evidence, we construct an independent source-native replay from public institutional release histories. The initial replay uses archived U.S. Bureau of Economic Analysis GDP releases and NOAA Climate Prediction Center ENSO discussions (U.S. Bureau of Economic Analysis, 2026; NOAA Climate Prediction Center, 2026). It freezes 23 endpoints before model outcomes are opened. The source-native experiment keeps the registered TimeSage mechanism families while maintaining a separate benchmark identity.

The initial evidence reveals distinct regimes. Documentary grounding produces a direct primary-track repair on DeepSeek and a held-out same-domain gain that also appears in the Kimi support-repair replication. Release alignment improves held-out Kimi performance when that cell moves away from ceiling. Cutoff filtering is initially ceiling-limited for the primary DeepSeek tasks, while an exploratory mini-model stress exposes a large cutoff effect on other endpoints.

Independent Mock-PC review then identifies institutional breadth as a decision-critical weakness. Before opening any new validation outcome, we freeze a post-review expansion from the EIA Weekly Petroleum Status Report archive (U.S. Energy Information Administration, 2026). It adds a third institutional source system, a third domain, and 12 new energy cutoff endpoints. The targeted and generic cutoff functions are reused byte-for-byte. On eight held-out April–May EIA endpoints, DeepSeek improves from 47.5% without the skill to 100% with targeted cutoff filtering. The same EIA tasks place the mini model at no-skill ceiling. The combined evidence therefore rejects a simple model-size account. A reusable procedure becomes causally important when the specific model-task regime leaves that operation unresolved.

This mechanism view changes how self-evolution should be evaluated. A recurring error can motivate a candidate state update. The update earns its place through controlled future intervention over a matched helper and the original agent. Transfer must then be tested on held-out temporal endpoints, with the procedure left unchanged. Our intervention identifies the value of an operation implemented as a callable skill; it does not establish that a skill wrapper is superior to implementing the same temporal operation upstream in retrieval or context construction.

Contribution and closest-work boundary. We operationalize an intervention-defined *procedure-bottleneck estimand*: for a predeclared recurring failure, a frozen targeted operation counts as a binding repair only when it beats both a complexity-matched target-blind helper and the original no-skill agent, after which unchanged-code transfer is adjudicated against ceiling and incompatible-domain nulls across BEA, NOAA, and EIA. The three-arm criterion is an identification contract, not a new skill-learning algorithm. Existing work already compiles reusable procedures from external knowledge and combines procedural skills with RAG; our claim is therefore neither skill acquisition nor RAG–skill complementarity, and it does not assert that the callable-skill wrapper is superior to an equivalent retrieval-side temporal implementation.

2 Recurring Failure as a Missing Procedure

Let a scenario generate a sequence of release periods $t_1, \ldots, t_K$. At each period, the agent receives a cutoff-valid evidence state E_t and a question x_t. The agent also carries a reusable skill library L_t.

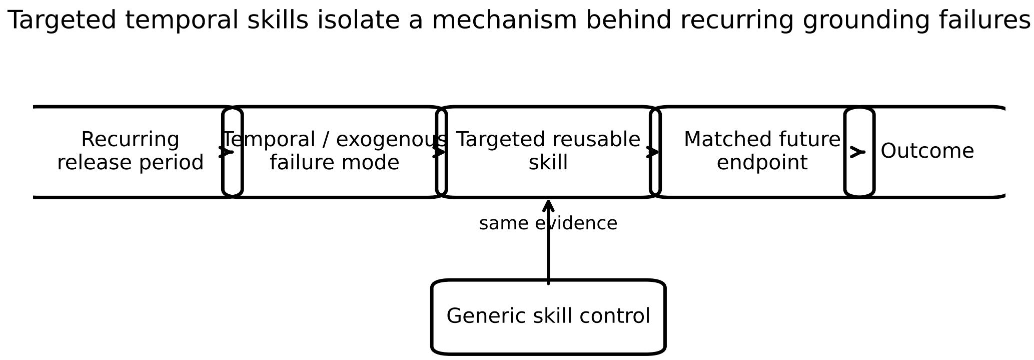

Figure 1: Targeted skill intervention. The same future endpoint is evaluated with a mechanism-specific skill and with a matched generic skill. This isolates the contribution of the procedure encoded by the targeted skill.

Future behavior follows

$$Y_t = H(x_t, E_t, L_t, S_t), \tag{1}$$

where S_t contains the remaining agent state.

A recurring failure mode appears when the same procedural error repeats across periods while the concrete evidence changes. This pattern suggests that the agent lacks a reusable operation. We represent the missing operation as a skill g_m associated with mechanism m.

The causal estimand compares outcomes on the same future endpoint under two skill libraries:

$$\Delta_m = \mathbb{E}[Y_t \mid do(L_t = L \cup \{g_m\})] - \mathbb{E}[Y_t \mid do(L_t = L \cup \{g_c\})], \tag{2}$$

where g_c is a generic control skill with matched implementation complexity. A positive Δ_m supports the claim that the target procedure repairs the matched failure beyond generic skill assistance.

2.1 Temporal cutoff skill

The cutoff skill receives a query date and a set of candidate evidence timestamps. It returns the admissible evidence subset under the benchmark cutoff. Its purpose is to prevent later releases from leaking into earlier-period analysis. The skill can also expose the cutoff date explicitly to subsequent reasoning.

2.2 Release alignment skill

The alignment skill maps source-release dates to the benchmark period represented by the evidence. It is useful when the source publication date differs from the underlying reporting period. The skill returns a normalized period identifier and verifies that the selected time-series row matches that period.

2.3 Exogenous grounding skill

The grounding skill links a numerical change to documentary evidence available before the cutoff. It searches prior-period reports for candidate events and returns evidence spans tied to the target period. This procedure supports attribution questions that require context outside the numerical series.

Each skill encodes a reusable operation that excludes scenario-specific answers. The same function can apply across future periods. This property matches the source harness contract for skills saved through `induct_skill`.

3 Source Evidence and Source-Native Replay

TimeSage-EV evaluates recurring analysis over institutional releases from February 2023 through May 2026 (Yao et al., 2026). It reports persistent temporal-validity gaps and recurring difficulty with exogenous context. TimeSage-1.0 adds a persistent skill library whose accepted procedures become callable on later periods. Its released aggregate results show higher performance and a token-cost ratio of 0.82$\times$ the reference condition.

The source benchmark also exposes a useful state-update boundary. `induct_skill` stores a reusable function. `retrieve_skill` exposes the stored implementation. Accepted skills become callable beginning with the next period. This chronological registration motivates our intervention contract.

3.1 Initial institutional replay

The initial replay is built directly from first-party release histories. Five BEA GDP releases cover the advance, second, and third estimates around 2025 Q4 and 2026 Q1 (U.S. Bureau of Economic Analysis, 2026). Five NOAA-CPC ENSO Diagnostic Discussions cover January through May 2026 (NOAA Climate Prediction Center, 2026). Raw pages are stored with SHA-256 hashes. Evidence identifiers are opaque content-derived IDs, so the identifier itself does not reveal the answer period.

Twenty-three endpoints are frozen before source-native model outcomes are opened. Twelve pre-April endpoints form C3-R. Eleven April–May endpoints form held-out C4-R. The three registered families retain their original success semantics. Cutoff validity checks whether cited evidence is admissible at the target date. Release alignment checks the represented reporting period together with the aligned headline value. Exogenous grounding checks first-party documentary spans against frozen evidence slots.

Grounding endpoints expose the full content-region sentence pool. Gold-only candidate sets are never exposed. The ten source documents contain between 19 and 90 candidate spans, with gold spans occupying only a small subset. The deterministic scorer uses no LLM judge. A pre-execution self-test gives 23/23 success on gold outputs and 0/23 on constructed negative outputs.

3.2 Post-review EIA expansion

After independent review raises source-breadth concerns, we freeze an additional cutoff-only validation before opening any EIA model outcome. EIA is selected by a structural compatibility rule: its Weekly Petroleum Status Report archive provides recurring first-party releases with explicit publication dates, represented data-ending weeks, and a cutoff operation executable by the already frozen T1/G1 interface (U.S. Energy Information Administration, 2026). The selection is therefore informed post-review validation, not a preregistered independent confirmation. No EIA model outcome is inspected during source selection, endpoint construction, scorer definition, or condition-order freezing. Sixteen archived pages yield twelve new energy endpoints. Each endpoint contains historical admissible releases plus exactly one next-release future distractor. Four endpoints form an earlier diagnostic split. Eight April–May endpoints form held-out validation.

The EIA expansion does not redesign the intervention. The targeted T1 cutoff function has the same source hash as in the initial replay. The matched G1 helper also remains byte-identical. Evidence IDs remain opaque. The EIA source manifest, endpoint set, prompt harness, scorer, and condition order are all content-addressed before the 360 post-review validation rows are executed.

3.3 Evidence layers

The registered primary model is DeepSeek-V4-Pro and completes five repeats. A second-model support-repair rule selects Kimi-K3 from the paper's pre-existing Mock-PC panel before any Kimi source-native task outcome is opened; Kimi is reported as a replication layer. A later provenance audit excludes the entire Kimi repeat-4 block after detecting that a controller retry overwrote that file, leaving four provenance-clean Kimi repeats. The primary-plus-replication initial replay therefore contains 621 runtime-valid rows. Doubao-Seed-2.0-Mini is a separately labeled exploratory capacity stress with 345 runtime-valid rows. The post-review EIA validation adds 360 runtime-valid rows across DeepSeek and the mini model. Across these retained evidence layers, the source-native study contains 1,326 runtime-valid rows over 35 distinct endpoints from three first-party institutional systems.

Each targeted skill has a family-matched generic helper with the same callable signature. In the frozen implementations, every targeted/generic pair has exactly equal Python lexical-token count and exactly equal AST-node count. External tool-call budgets are equal. Static audits reject generic helpers that access the target operation.

4 TARGETED SKILL INTERVENTION

Each frozen endpoint is replayed under three conditions while its evidence package, model configuration, prompt harness, evaluator, and period remain fixed.

The targeted condition injects the registered mechanism-specific procedure. The generic condition injects a frozen target-blind helper under the same callable wrapper. The no-skill condition adds no experimental helper. We report targeted minus generic and targeted minus no skill together. This joint interpretation separates genuine repair from a contrast created by generic-helper degradation.

4.1 MECHANISM CONTRACTS

The cutoff skill filters candidate evidence by the target cutoff. Its generic control validates package structure without using timestamps. The release-alignment skill normalizes the reporting period represented by a source release. Its generic control inspects document structure without performing date-to-period mapping. The grounding skill enriches candidate documentary spans using reusable attribution and outlook cues. Its control can expose document structure but has no contextual event selector.

No reusable skill contains scenario-specific values, dates, labels, or expected answers. Skill source hashes are frozen before execution. The targeted and generic source implementations are matched exactly in lexical-token count and AST-node count for each family.

4.2 FROZEN FAMILY OUTCOMES AND SCORER TIMELINE

Before any source-native model outcome is opened, the failure-family contracts define what constitutes success. A cutoff success requires a nonempty set of cited evidence and a selected latest reference that are all available by the target cutoff. A release-alignment success requires the correct reporting-period identity together with the aligned headline value. A grounding success requires the requested evidence slots to be filled by frozen first-party sentence IDs in the correct slot order.

The r3 implementation additionally stores a full-task conjunctive exact scorer. It is a different estimand, not a more conservative version of the family estimand: cutoff success under the registered contract concerns temporal admissibility, while the conjunctive score also requires exact reporting-period and headline-value fields; alignment success concerns period identity and aligned value, while the conjunctive score additionally requires exact surface-form fields such as release-stage strings. After outcomes are opened, the audit shows that these extra conjuncts can drive the full-task score to zero even when the registered mechanism is corrected. We preserve every raw composite score and report it as sensitivity analysis. Primary analysis uses the family definitions frozen before execution. Period canonicalization maps only semantically equivalent quarter strings to the same period identity and never changes endpoints, predictions, gold values, or the intervention. The EIA validation removes this representation ambiguity by freezing its cutoff family scorer before any r4 outcome.

The causal contrasts are

$$\Delta_{T-G} = S_{\text{target}} - S_{\text{generic}}, \qquad \Delta_{T-N} = S_{\text{target}} - S_{\text{no skill}}. \tag{3}$$

A positive Δ_{T-G} with a nonpositive Δ_{T-N} is not counted as repair. This pattern indicates that the generic helper may have degraded behavior while the targeted condition merely restored the original baseline.

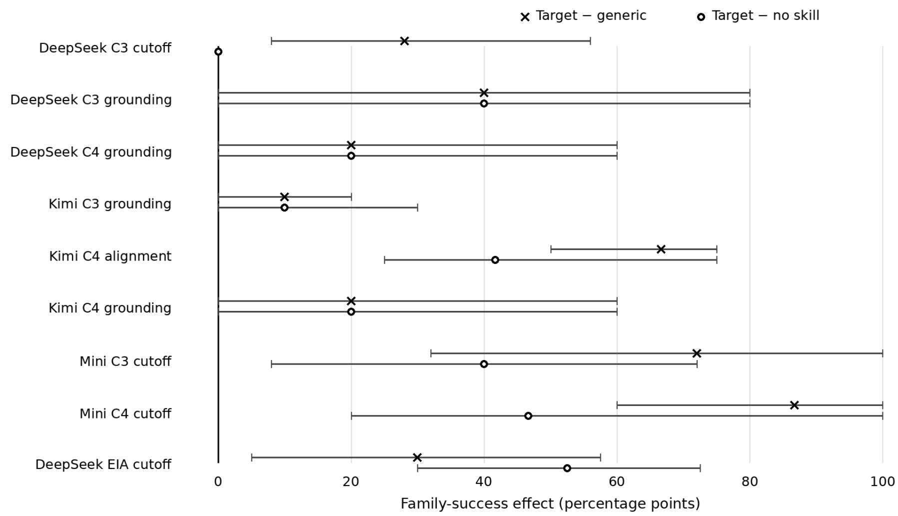

Figure 2: Temporal-skill effects are regime-dependent. Circles compare the targeted skill with the no-skill agent; crosses compare it with the matched generic helper. Horizontal bars show endpoint-bootstrap 95% intervals. A targeted-minus-generic gain alone is insufficient: DeepSeek C3 cutoff has a positive two-arm contrast but zero repair over no skill. EIA denotes the frozen post-review energy validation.

4.3 Chronological split and transfer

C3-R uses the twelve pre-April endpoints from the initial replay. C4-R contains eleven held-out April–May endpoints. The skill source is unchanged across the split. Same-domain and compatible cross-domain transfer labels are frozen in the initial source-native schedule before outcomes. The post-review EIA expansion independently freezes four earlier and eight held-out energy endpoints while retaining T1 and G1 unchanged.

4.4 Execution hierarchy

DeepSeek is the registered primary scientific track. Its first repeat acts as the F0 falsifier; later repeats execute only after runtime-validity and structural-degeneracy gates pass. The secondary-model repair rule is also fixed before any replacement-model task outcome: if the planned local secondary falls below 90% runtime validity, select the only non-primary model in the paper's pre-r3 post-repair Mock-PC panel. That panel contains DeepSeek and Kimi, while DeepSeek is already primary, so the rule uniquely resolves to Kimi-K3. Only the model identifier changes; the same 69 F0 slots and frozen condition order are reused. Kimi is therefore a separately labeled replication. The mini-model layer is exploratory capacity stress. The EIA layer is post-review targeted validation prompted by independent review and frozen before its own model outcomes.

Every result row binds the endpoint, cutoff, failure family, condition, repeat, skill hash, prompt hash, raw model response hash, and provider response identifier hash. This binding prevents later interpretation from drifting away from the executed experimental object.

5 Results

Table 1 reports the frozen family-success criteria for the registered DeepSeek primary track and the Kimi support-repair replication. DeepSeek retains five repeats and Kimi retains four provenance-clean repeats. Repeats measure execution robustness; they are not treated as independent task samples. Each rate pools endpoint-repeat observations within the displayed cell. Figure 2 shows the endpoint-level causal contrasts for the non-ceiling cells and the DeepSeek cutoff false-positive control case.

Model / split	Family	N0	Generic	Target	Δ_{T-G}	Δ_{T-N}
DeepSeek C3-R	Cutoff	100	72	100	+28	0
DeepSeek C3-R	Align	100	100	100	0	0
DeepSeek C3-R	Ground	20	20	60	**+40**	**+40**
DeepSeek C4-R	Cutoff	100	87	100	+13	0
DeepSeek C4-R	Align	100	100	100	0	0
DeepSeek C4-R	Ground	40	40	60	**+20**	**+20**
Kimi repl. C3-R	Cutoff	100	100	100	0	0
Kimi repl. C3-R	Align	100	100	100	0	0
Kimi repl. C3-R	Ground	30	30	40	+10	+10
Kimi repl. C4-R	Cutoff	100	100	100	0	0
Kimi repl. C4-R	Align	58	33	100	**+67**	**+42**
Kimi repl. C4-R	Ground	60	60	80	**+20**	**+20**

Table 1: Primary and support-repair replication family success (percent). DeepSeek has five repeats; Kimi retains four provenance-clean repeats. Bold contrasts improve over both controls.

5.1 Grounding gives direct primary repair and future replication

DeepSeek shows a stable acquisition-side grounding effect. Across 25 endpoint-repeat observations per condition, N0 and G3 each succeed on 5 while T3 succeeds on 15. Thus $\Delta_{T-G} = \Delta_{T-N} = +0.40$, and the direction is unchanged across all five repeats. Kimi replication shows a smaller C3-R gain of 10 points over both controls across its four retained repeats.

The held-out grounding result is directionally consistent across the two evidence layers. DeepSeek moves from 40% under either control to 60% with T3. Kimi moves from 60% to 80%. Both gains are 20 points. On the registered same-domain held-out subset, both models gain 50 points over G3 and N0. Cross-domain grounding shows no incremental targeted effect. The current grounding procedure therefore transfers across future periods more reliably than across the registered domain boundary.

Endpoint-level uncertainty makes the small original replay explicit. DeepSeek C3-R grounding has mean $\Delta_{T-G} = 40$ points with an endpoint-bootstrap 95% interval of [0,80] and two wins, three ties, no losses. DeepSeek C4-R grounding is +20 points with [0,60] and one win, four ties. The four-repeat Kimi replication gives +10 points on C3-R grounding with [0,20], then +20 points on C4-R with [0,60]. The registered same-domain grounding subset contains only two held-out endpoints, so its 50-point contrast is descriptive evidence of future-period transfer and is not presented as a population-level significance result.

5.2 The no-skill anchor rejects a false cutoff repair

DeepSeek C3-R appears positive under a two-arm comparison: T1 is 100% and G1 is 72%. N0 is also 100%. The 28-point Δ_{T-G} is therefore caused by generic-helper degradation and provides no repair over the original agent. The mechanism is visible in the frozen G1 code: it validates schema-compatible evidence and can return both the admissible release and the post-cutoff distractor because it is intentionally forbidden from comparing release dates with the cutoff. This structurally useful yet temporally blind output can foreground future evidence. T1 removes that ambiguity by applying the registered cutoff operation. The same pattern appears weakly in C4-R, while Kimi is at ceiling. A matched generic helper alone is insufficient for causal interpretation; the no-skill anchor changes the scientific verdict.

5.3 Cutoff repair depends on the model-task regime

The exploratory mini-model stress exposes a non-ceiling cutoff regime in the initial replay. Table 2 shows five-repeat results. In C3-R, T1 reaches 72%, compared with 32% for N0 and 0% for G1. In held-out C4-R, T1 reaches 100%, compared with 53% for N0 and 13% for G1. The held-out cross-domain cutoff subset is 0/5 for N0, 0/5 for G1, and 5/5 for T1.

The post-review EIA validation shows that the same mechanism is not monotonic in model size. We freeze 12 new energy cutoff endpoints before opening r4 outcomes and reuse T1/G1 unchanged. On

Split	Family	N0	Generic	Target	Δ_{T-G}	Δ_{T-N}
C3-R	Cutoff	32	0	72	**+72**	**+40**
C3-R	Align	100	100	100	0	0
C3-R	Ground	40	48	64	+16	+24
C4-R	Cutoff	53	13	100	**+87**	**+47**
C4-R	Align	80	93	100	+7	+20
C4-R	Ground	60	52	64	+12	+4

Table 2: Exploratory mini-model capacity stress over five repeats (percent). This evidence layer is reported separately from the primary and replication tracks.

the eight held-out April–May endpoints, DeepSeek moves from 19/40 under N0 and 28/40 under G1 to 40/40 under T1. This is +52.5 points over no skill and +30 points over the generic helper. The targeted-minus-no-skill contrast is positive on seven of eight endpoint means, with one tie and no losses. An endpoint-resampled percentile bootstrap gives a 95% interval of [30, 72.5] points; an exact two-sided sign test on the seven non-tied endpoints gives $p = 0.0156$. The targeted-minus-generic endpoint interval is [5, 57.5] points, with four wins and four ties; its sign test gives $p = 0.125$.

The mini model is already at 40/40 no-skill success on the same held-out EIA endpoints, so T1 cannot improve its baseline there. The combined result rejects a simple "smaller model, larger skill effect" explanation. A cutoff procedure becomes a bottleneck when a particular model-task pair leaves temporal filtering unresolved.

5.4 Release alignment is selective

C3-R alignment is at ceiling in every evaluated initial-replay model, so that split does not identify a causal effect. Held-out Kimi moves away from ceiling: T2 succeeds on 12/12 observations, compared with 7/12 for N0 and 4/12 for G2. The resulting gains are +41.7 points over no skill and +66.7 points over the generic helper. At the endpoint level, targeted-minus-generic is positive on all three held-out alignment endpoints, with a percentile bootstrap interval of [50, 75] points; the exact sign test is $p = 0.25$ because the endpoint count is three. DeepSeek remains at ceiling. The current evidence supports a reusable alignment safeguard in one non-ceiling model and does not establish a model-invariant effect.

Taken together, the evidence supports a conditional mechanism claim. Reusable temporal procedures become causal bottlenecks when the evaluated model-task regime lacks the corresponding operation. Ceiling performance can hide the intervention effect. Successful future transfer can also remain narrower than the apparent abstraction level of the procedure.

6 Related Work

Skill attribution, compilation, and evolution. Paired trace auditing, no-skill/matched-reference attribution, randomized masking, and continual skill evaluation already establish heterogeneous skill effects (Zhou et al., 2026; Dong et al., 2026; Wang et al., 2026b; Guan et al., 2026). Evo-Harness, SkillProx, and HyperSkill further compile, optimize, or structure reusable skills (Wei et al., 2026; Zheng et al., 2026; Xu et al., 2026). Anything2Skill goes further by compiling heterogeneous external knowledge into persistent procedural skills and retrieving those skills together with RAG passages (Pan et al., 2026). Reusable-procedure compilation and RAG–skill complementarity are therefore not our novelty. The residual object is whether a *predeclared recurring failure* is a procedure bottleneck: the same frozen operation must repair it over both generic assistance and the original agent, and then survive explicit transfer/ceiling adjudication.

Retrieval as an alternative treatment surface. Self-RAG and CRAG provide generic retrieval/self-correction baselines (Asai et al., 2024; Yan et al., 2024), while Anything2Skill demonstrates that RAG passages and reusable procedural skills can be complementary in one deployed agent (Pan et al., 2026). Temporal systems place cutoff/release reasoning upstream in retrieval or temporal structure (Han et al., 2025; Wang et al., 2026a; Zhang et al., 2025; Abdallah et al., 2026; Liu et al., 2026). These works already cover temporal filtering and RAG–procedure composition. We therefore claim only operation-level bottleneck identification under our three-arm intervention, not uniqueness or superiority of a callable skill container over an equivalent retrieval-side implementation.

7 LIMITATIONS

The exact TimeSage-EV May 2026 evaluated snapshot remains unavailable from the declared first-party release points as of the 2026-08-22 support audit: the audited GitHub revision exposes only its README and the Hugging Face revision only `.gitattributes`. Our source-native replay is independent mechanistic evidence, not an exact TimeSage-EV rerun. The frozen TimeSage C3/C4 replication remains open until a hashable evaluated object exposes the required inputs, evaluator contract, temporal metadata, and executable skill interface.

The source-native study contains 35 endpoints across three first-party institutional systems and three domains, with the post-review expansion concentrated on cutoff reasoning. Grounding and release alignment still rely on the original BEA/NOAA set. The same-domain grounding transfer subset has only two held-out endpoints, and cross-domain grounding shows no incremental targeted effect. We therefore keep broad grounding portability outside the supported claim. The EIA expansion strengthens one procedure family and does not provide a new-domain replication for all three mechanisms.

The planned local Qwen secondary fails the 90% runtime-validity gate after twelve attempts, and no prompt or parser repair is made. The fixed support-repair rule then uniquely resolves to Kimi before any Kimi source-native outcome. A later hash audit shows that an MCP 502 retry overwrote only Kimi repeat 4; we exclude that whole repeat block and run no replacement, leaving four provenance-clean Kimi repeats. The mini-model stress is exploratory, and the EIA expansion is post-review validation; both are frozen before their own outcomes.

The r3 harness also stores an auxiliary full-task exact scorer. A post-outcome audit finds representation penalties outside the pre-execution family definitions, so raw strict scores are preserved while the registered family contracts remain primary. The EIA family scorer is frozen before r4. Repeated executions measure robustness and are not independent task samples. We report endpoint-level bootstrap intervals and sign tests where endpoint counts permit them, and treat the smallest transfer subsets descriptively.

The generic helpers are matched in callable signature, lexical-token count, AST-node count, and external tool budget, but they are not guaranteed to be behaviorally neutral. The no-skill arm is therefore part of the identification design: it prevents generic-helper degradation from being misread as targeted repair, as the DeepSeek cutoff false positive demonstrates. We have not evaluated a distribution of alternative behavior-neutral generic controls, so targeted-minus-generic effect size should not be interpreted as invariant to the choice of helper. Likewise, we do not compare against an equivalent retrieval-side temporal prefilter or timestamp-aware retriever. Such controls would separate the value of the temporal operation from the particular choice to package it as callable agent state.

Condition schedules are frozen before outcomes and vary condition position across endpoint/model slots; the EIA builder deterministically shuffles the three arms from a frozen seed. A zero-call audit over all 1,326 retained runtime-valid rows confirms that helper execution is harness-forced and identical in rate for targeted and generic arms (442/442 each), while no-skill executes no helper (0/442); pooled success has no detectable position association in any arm ($p > .65$). Targeted prompts are also shorter than their exact generic counterparts by 18.85 input tokens on average. These checks reduce call-rate, prompt-length, and fixed-position explanations, but they are post-hoc robustness diagnostics rather than an independently counterbalanced experiment.

8 CONCLUSION

Reusable temporal skills become intervention-defined causal bottlenecks in specific non-ceiling model-task regimes. Here bottleneck means that the frozen targeted procedure repairs the registered mechanism over both matched generic assistance and the original agent; it does not assert that no alternative intervention could also repair the failure. Documentary grounding produces direct DeepSeek repair and future same-domain replication. Cutoff filtering produces large repair in the exploratory mini initial replay and, with the same T1 code, in the post-review DeepSeek EIA validation; the mini model is simultaneously at EIA ceiling, rejecting a monotonic model-size explanation. Release alignment improves held-out Kimi performance when that cell leaves ceiling. The no-skill anchor also exposes cases where a generic helper creates a false targeted-minus-generic signal.

A recurring failure can therefore identify a candidate persistent procedure. Promotion to agent state should depend on controlled future evidence against both generic assistance and the original agent.

Reproducibility Statement

The source-native study is content-addressed end to end. The supplementary replication bundle contains the frozen endpoint definitions, prompt harnesses, family scorers, all six executed targeted/generic skill implementations, condition schedules, first-party archived source pages, retained raw result JSON, adjudication code, and a SHA-256 manifest. Appendix A records the executed helper code and the endpoint-level inferential protocol. The source-native bundle is independent of the unavailable TimeSage-EV evaluated snapshot; the exact TimeSage replication remains a separately identified external replication target.

AI Use Statement

Generative AI tools were used throughout the research workflow to assist literature triage, research-idea and hypothesis refinement, experimental-design critique, implementation and debugging, data reformatting, result interpretation, internal mock review, and manuscript drafting and editing. AI-assisted experimental artifacts were accepted only after executable checks against frozen task contracts, immutable raw outputs, first-party source material, or deterministic recomputation. AI-generated review feedback had advisory status and did not itself create scientific evidence. The authors reviewed the AI-assisted work and take responsibility for the final paper, claims, code, and supplementary artifacts.

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A Reproducibility and Sensitivity Audits

T1/G1 are byte-identical across the initial replay and EIA validation. The provenance-clean r3 readuit is `6221016166b71ebe`; r4 adjudication is `316ee89a8eb1d53f`. The r5 replication bundle contains executable r3/r4 endpoints, harnesses, scorers, runner scripts, frozen skill source, first-party archived pages, retained result JSON, and a SHA-256 manifest. This bundle reproduces the source-native study and remains separate from the unavailable TimeSage-EV evaluated snapshot.

A.1 Exact frozen reusable helpers

The six helper implementations below are the executed source, not pseudocode. Each targeted/generic pair has equal lexical-token count and equal AST-node count under the pre-execution audit.

```
# T1_temporal_cutoff.py
def skill(package, context):
    limit = context.get("cutoff_date", "")
    chosen = []
    for item in package.get("evidence_metadata", []):
        marker = item.get("release_date", "")
        ref = item.get("evidence_ref", "")
        if isinstance(marker, str) and marker <= limit:
            chosen.append(ref)
    return {"evidence_refs": chosen}
```

```python
# G1_temporal_cutoff.py
def skill(package, context):
    limit = context.get("schema_ceiling", "")
    chosen = []
    for item in package.get("evidence_metadata", []):
        marker = item.get("schema_version", "")
        ref = item.get("evidence_ref", "")
        if isinstance(marker, str) and marker <= limit:
            chosen.append(ref)
    return {"evidence_refs": chosen}

# T2_release_alignment.py
import re
def skill(package, context):
    text = " ".join(package.get("document_text", "").split())
    match = re.search(r"([1-4])(?:st|nd|rd|th) Quarter(?: and Year)? (20\d{2})",
                      text, re.I)
    value = None
    if match:
        value = f"{match.group(2)}-Q{match.group(1)}"
    return {"value": value}

# G2_release_alignment.py
import re
def skill(package, context):
    text = " ".join(package.get("document_text", "").split())
    match = re.search(r"(News) (Release|Archive|Statement)(?: and Data)? "
                      r"([A-Za-z]{2,20})", text, re.I)
    value = None
    if match:
        value = f"{match.group(1)}-{match.group(3)}"
    return {"value": value}

# T3_exogenous_grounding.py
def skill(package, context):
    markers = ("contributors to", "reflected", "because", "collectively",
               "favored", "favors ", "associated with", "due to")
    chosen = []
    for span in package.get("candidate_spans", []):
        text = span.get("text", "").lower()
        sid = span.get("span_id", "")
        if any(marker in text for marker in markers):
            chosen.append(sid)
    return {"span_ids": chosen}

# G3_exogenous_grounding.py
def skill(package, context):
    markers = ("table", "figure", "release", "appendix", "contact",
               "notes ", "technical notes", "page")
    chosen = []
    for span in package.get("candidate_spans", []):
        text = span.get("text", "").lower()
        sid = span.get("span_id", "")
        if any(marker in text for marker in markers):
            chosen.append(sid)
    return {"span_ids": chosen}
```

A.2 Endpoint-level uncertainty and test resolution

Table 3 averages repeats within endpoint-condition and bootstraps endpoints; W/T/L is targeted versus generic, and sign tests use non-ties. Repeats never increase the inferential sample size. For n non-tied endpoints, the smallest attainable two-sided sign-test value is 2^{1-n} when every endpoint moves in the same direction. Thus two non-tied endpoints cannot yield a value below .5; three cannot go below .25; five cannot go below .0625; eight can reach .0078. Ties reduce resolution further. This arithmetic is why the two-endpoint grounding-transfer result is explicitly descriptive, why the three-endpoint Kimi alignment result remains underpowered despite a large effect, and why the eight-endpoint EIA validation supports a stronger endpoint-level claim.

A.3 Why the family estimand is primary

The family contract and the full-task conjunctive score answer different questions. The family score asks whether the targeted temporal operation was corrected. The conjunctive score additionally requires orthogonal output fields and exact surface forms. A zero conjunctive score can therefore coexist with a successful cutoff intervention when temporal admissibility is correct but an unrelated

Layer	Split	Fam.	Δ_{T-G} [95%]	W/T/L	p_G	Δ_{T-N} [95%]	p_N
D	C3	Cut	28 [8,56]	4/1/0	.125	0 [0,0]	1
D	C3	Grd	40 [0,80]	2/3/0	.50	40 [0,80]	.50
D	C4	Cut	13 [0,20]	2/1/0	.50	0 [0,0]	1
D	C4	Grd	20 [0,60]	1/4/0	1	20 [0,60]	1
K	C3	Grd	10 [0,20]	2/3/0	.50	10 [0,30]	1
K	C4	Align	66.7 [50,75]	3/0/0	.25	41.7 [25,75]	.25
K	C4	Grd	20 [0,60]	1/4/0	1	20 [0,60]	1
M	C3	Cut	72 [32,100]	4/1/0	.125	40 [8,72]	.25
M	C3	Grd	16 [-12,60]	1/3/1	1	24 [0,64]	.50
M	C4	Cut	86.7 [60,100]	3/0/0	.25	46.7 [20,100]	.25
M	C4	Align	6.7 [0,20]	1/2/0	1	20 [0,60]	1
M	C4	Grd	12 [0,28]	2/3/0	.50	4 [-32,36]	1

Table 3: Endpoint-level uncertainty for non-ceiling initial-replay cells. D=DeepSeek primary, K=Kimi replication, M=exploratory mini.

field differs, or with successful release alignment when a semantically identical period is formatted differently. The family definitions were frozen before execution, so selecting them is determined by the experimental question and its time order. The conjunctive score is preserved as a sensitivity view and is never used to erase an unfavorable family result.

Family	r3 full-task conjunctive score	Effect on claim
Grounding	Identical to family score	Unchanged
Cutoff	All cells collapse to 0	Family estimand remains primary
Alignment	All cells collapse to 0	Family estimand remains primary

Table 4: Sensitivity to the auxiliary full-task exact score. The two columns evaluate different estimands.

A.4 CALL-RATE, PROMPT-LENGTH, AND CONDITION-POSITION AUDIT

The retained source-native rows contain 442 no-skill, 442 generic, and 442 targeted observations. Helper execution is deterministic in the frozen harness: generic and targeted execute exactly once before the model answer in 442/442 rows each, whereas no-skill executes no helper in 0/442 rows. Thus targeted-minus-generic cannot be explained by a higher helper invocation rate; this experiment does not measure endogenous tool-use propensity. Across exact targeted/generic pairs, targeted prompts are shorter by 18.85 input tokens on average (302 shorter, 140 longer). Condition-position counts are 157/135/150 for no-skill, 145/155/142 for generic, and 140/152/150 for targeted across positions 0/1/2. Pooled 3×2 success-by-position tests give $p = .993$, $.657$, and $.914$ for no-skill, generic, and targeted respectively; targeted-minus-no-skill remains positive at all three positions. These are post-hoc zero-call diagnostics and do not substitute for a prospectively counterbalanced replication. The full audit is included in the supplement.

Every retained row binds endpoint, condition, repeat, skill hash, prompt hash, raw response hash, and provider response-ID hash. Kimi repeat 4 is excluded by hash provenance with no replacement. The exact TimeSage-EV replication remains separate from all source-native evidence.